

Ray-Yuan (Ray) Chung

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A human-centered AI researcher with a background in behavioral science and health informatics

Research Interests: Human-AI Interaction, Health AI, Explainable AI, Agentic AI

EDUCATION

University of Washington, Seattle, WA 2023–2027 (anticipated)
Ph.D. in Biomedical and Health Informatics, Concentration: HCI and AI/ML
University of Michigan, Ann Arbor, MI 2018–2020
M.P.H. in Nutritional Sciences and Dietetics
National Chung-Hsing University, Taichung, Taiwan 2013–2017
B.S. , Concentration: Pre-Med

RELEVANT EXPERIENCE

University of Washington, Seattle, WA Sep 2023–Present
Graduate Researcher

- *Mental Health Knowledge Graph*: Built a multi-agent-based mental health knowledge graph and QA dataset; assessed efficacy via automated and human evaluations.
- *Co-Design AI Decision-making Intervention* : Co-designed an AI-powered decision-support tool with kidney patients, caregivers, and clinicians through surveys and vignette-based values elicitation.
- *AI Chatbot for Older Adults*: Developed and evaluated a chatbot with accessible features for older adults using OpenAI API and Streamlit.
- *Online Health Misinformation*: Applied mixed-effects regression and LLaMA-based text classification (few-shot + CoT prompting) to study user engagement and perception of health misinformation on YouTube.
- *Health Device Usability*: Conducted interviews and surveys with cystic fibrosis patients to evaluate the usability of an at-home lung monitoring device.

Unita Health, Santa Clara, CA (Remote) Sep 2024–Sep 2025
Human-AI Interaction Researcher

- Led end-to-end development of an LLM conversational agent for gastrointestinal health from building a RAG pipeline to usability evaluation.

Dexcom, San Diego, CA Jun 2024–Sep 2024
User Experience Research Intern

- Conducted mixed-methods user research (TTM-informed surveys and interviews) to inform CGM product development and behavioral feature design.

Impossible Foods, Redwood City, CA May 2022–Aug 2023
Research Scientist

- Translated analysis of large-scale ingredient and nutrient datasets into actionable insights that informed product innovation and business strategy.

Heali AI, Remote Mar 2021–May 2021
Nutrition AI Researcher

- Curated and annotated nutrition datasets to train an AI menu scanner for personalized dietary insights.

PUBLICATIONS

- **Ray-Yuan Chung**, Jaime Snyder, Zixuan Xu, Daeun Yoo, Athena C. Ortega, Wanda Pratt, Aaron Wightman, Ryan Hutson, Cozumel Pruette, Ari Pollack. Design Implications for Collaborative Decision-making Technologies in Pediatric Chronic Care. *CHI 2026 (Under Review)*.
- Feng Chen, Manas Satish Bedmutha, **Ray-Yuan Chung**, Janice Sabin, Wanda Pratt, Brian R Wood, Nadir Weibel, Andrea L Hartzler, Trevor Cohen. Toward Automated Detection of Biased Social Signals from the Content of Clinical Conversations. 2025. *AMIA Annual Symposium Proceedings*.
- Luna Xingyu Li, **Ray-Yuan Chung**, Feng Chen, Wenyu Zeng, Yein Jeon, Oleg Zaslavsky. Learning from Elders: Making an LLM-powered Chatbot for Retirement Communities more Accessible through User-centered Design. 2025. In *Proceedings of the CALD-AI Workshop at ASIS&T*.
- **Ray-Yuan Chung**, Feng Chen, Yein Jeon, Oleg Zaslavsky. Improve Technology and EHealth Literacy in a Retirement Community via the Use of LLM-Powered Chatbot. 2024. *Innovation in Aging*.
- Prasad P Devarshi, **Ray-Yuan Chung**, Carroll A Reider, Ryan W Grant, Susan Hazels Mitmesser. Nutrient Gaps in US Adults by Age and Gender: Vitamin A, D, E, K, C, Magnesium, Calcium, Choline and Dietary Fiber. 2020. *Journal of the Academy of Nutrition and Dietetics*.
- Carroll A Reider, **Ray-Yuan Chung**, Prasad P Devarshi, Ryan W Grant, Susan Hazels Mitmesser. Inadequacy of immune health nutrients: intakes in US adults, the 2005–2016 NHANES. 2020. *Nutrients*.

AWARDS

- Top Scholars Award, UW (2023)
- Global Scholars Award, NSF International (2018–2020)
- Maxine Moore Scholarship, UM SPH (2019)

SKILLS

Programming & Tools: Python, R, JavaScript, PyTorch, TensorFlow, Hugging Face, LangChain, Docker, Git, Tableau, RedCap, Qualtrics, Prolific

AI/ML: LLMs, Fine-tuning, Prompt Engineering, RAG, Multi-Agent Systems, NLP, Multimodal Learning, Reinforcement Learning

UX/HCI: Interviews, Focus Groups, Surveys, Usability Testing, A/B Testing, Regression Modeling, Inclusive Design, Participatory Design

Behavioral Science: Behavior Change Design, Health Behavior Theories

AFFILIATIONS / LICENSURES

- Registered Dietitian (2021 - present)
- The DUB Group at University of Washington